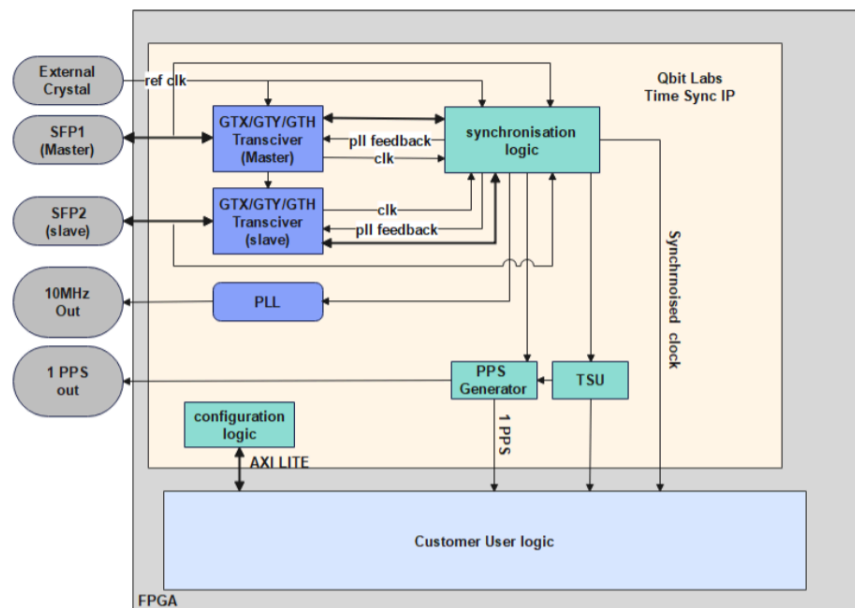


QbitLabs Time Sync IP with Daisy Chain Support (QWR-IP-SX-02)

The QbitLabs Time Sync IP with daisy chain support is an FPGA soft IP core designed to provide sub-nanosecond synchronization accuracy over 1 Gbps optical fiber links. It implements a White Rabbit (WR) boundary clocking on the FPGA, synchronizing to a compatible WR master.

The core achieves precise time and frequency distribution by combining Synchronous Ethernet, hardware timestamping, and dynamic phase/frequency corrections via internal PLL feedback and synchronization logic—all implemented using standard FPGA resources (no external dedicated timing hardware required).



Architecture

The White Rabbit Time Sync IP is a high-precision FPGA-based timing core designed to deliver sub-nanosecond time and frequency synchronization. At its core, the architecture features two independent GTX/GTY/GTH transceivers: one configured as Master (connected to SFP1) and the other as Slave (connected to SFP2). An external crystal provides a stable reference clock directly to the master transceiver. The IP fully supports daisy-chain topologies, allowing multiple nodes to be synchronized in series without requiring external Ethernet switches. Recovered clocks and PLL feedback signals from both transceivers are fed into the central synchronisation logic block, which implements the White Rabbit protocol for accurate phase, frequency, and time alignment. A dedicated dynamic calibration mechanism measures and compensates for fiber delays and link asymmetries in real time, ensuring long-term stability despite temperature and environmental variations. For dynamic calibration hw support is needed. A PLL block generates a clean 10 MHz output, while a PPS Generator produces a precise 1 Pulse Per Second signal locked to the grandmaster time base. The Time Stamp Unit (TSU) provides hardware-accurate timestamping for all White Rabbit packets and events. Configuration and status registers are accessible to the customer user logic through an AXI Lite interface. Synchronized clock, 1 PPS, and 10 MHz outputs are routed both externally and to the customer user logic, enabling seamless integration into larger FPGA designs. The architecture ensures low jitter and high timing stability across master and slave modes, making it ideal for building scalable, distributed, high-precision timing systems.

Why Choose Us?

 Rapid integration, faster time-to-market

 Clean, verified, documented RTL

 Direct access to experts



Features

- 2 ports node which can be configured to work as master (daisy chain mode) or slave mode interdependently. Number of ports can be configured to 1 or 2 in compile time.
- Sub-nanosecond synchronization over 1 Gbps optical fiber using White Rabbit protocol
- Sub-nanosecond time accuracy over 1 Gbps Ethernet optical fiber interfaces using White Rabbit (basis of IEEE 1588-2019 High Accuracy).
- Easy deployment: plug & play solution over 10 km using optical fiber links without need for link calibration and upto 100Kms with daisy chain topology.
- Dynamic compensation of asymmetries caused by environmental conditions (e.g., temperature).
- High-accurate and deterministic timing integration directly on the FPGA.
- No need for expensive external oscillators / clocks / VCXOs or dedicated hardware beyond standard FPGA transceivers.
- Optimized FPGA resource utilization.

Key Capabilities

- Sub-nanosecond time transfer and stable frequency dissemination over the same optical link.
- Internal timestamping with 8 ns/16ns granularity for TAI (International Atomic Time) counters.
- Outputs: 1 PPS signal, high-resolution timestamps, and a synchronized 10Mhz clock, synchronized reference clock for user logic.
- The IP integrates the full stack from Xilinx transceiver interface to timing outputs, with AXI4-Lite for configuration, monitoring, and status from user logic. No external CPU is required.

Supported Tools and Devices

- Xilinx Vivado 2023.x or later (IP integrator or direct HDL integration).
- Supported FPGA families: Kintex-7, Virtex-7, Artix-7, Zynq-7000, Zynq UltraScale+, and equivalent (GTX/GTY/GTH capable).
- Contact us for support on other families or tools.

Hardware Requirements

- Physical connection to an external SFP cage (via GTX/GTY/GTH transceiver pins).
- External reference clock input (recommended: 125 MHz or compatible for GT);
- One external crystal/oscillator for reference clock (ref clk).
- Redundancy is supported by instantiating two instances of this IP if (HW support is needed for external physical connections)
- For Dynamic Calibration, to utilize calibration state of WR state machine, HW support is needed to loop back the transceiver output back to the IP core.
- Supported on Xilinx 7-series and UltraScale(+) families with GT transceivers.

Use Cases & Applications

Typical applications in timing-critical infrastructures:

- Quantum communication and Quantum Networks for ultra precise timing synchronization.
- Distributed sensor arrays (e.g., radar, phased arrays) for defense and aerospace.
- High-frequency trading timestamping devices and financial network capture synchronization.
- Datacenter server/NIC synchronization for low-jitter, deterministic timing.
- Phased-array antennas, scientific instrumentation, and telecom backbone timing.

Deliverables

- Encrypted HDL source code, constraints, user guide, and license activation (per device/project).
- Evaluation/Permanent license from Qbit Labs.



Mode of Operation

- 1. Slave Mode (Model: QWR-IP-EP-02):** In this mode the ethernet ports work as WR end point Slave node . Users have flexibility to generate the IP for 1 Port or 2 Ports.
- 2. Daisy Chain Mode (Model: QWR-IP-SX-02):** In this mode, one port works as master node and other one as slave node. Timing is transferred from slave node to master node. It implements boundary clocking mode of WR protocol.

Performance

Typical measured performance (example; replace with your measurements):

- Time error (TE) vs. WR master 1 PPS: ± 200 ps peak-to-peak.
- Standard deviation: $\sim 40\text{--}50$ ps.
- Long-term stability with short fiber: mean offset ~ 50 ps, peak-to-peak ~ 280 ps.

FPGA Resource Utilization Example

LUT	Registers	BRAM	DSP	MMCM	GT
$\sim 10,000$	$\sim 8,000$	~ 68.5	5	2	3

Get in touch with us

Delivering Advanced Solutions for
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